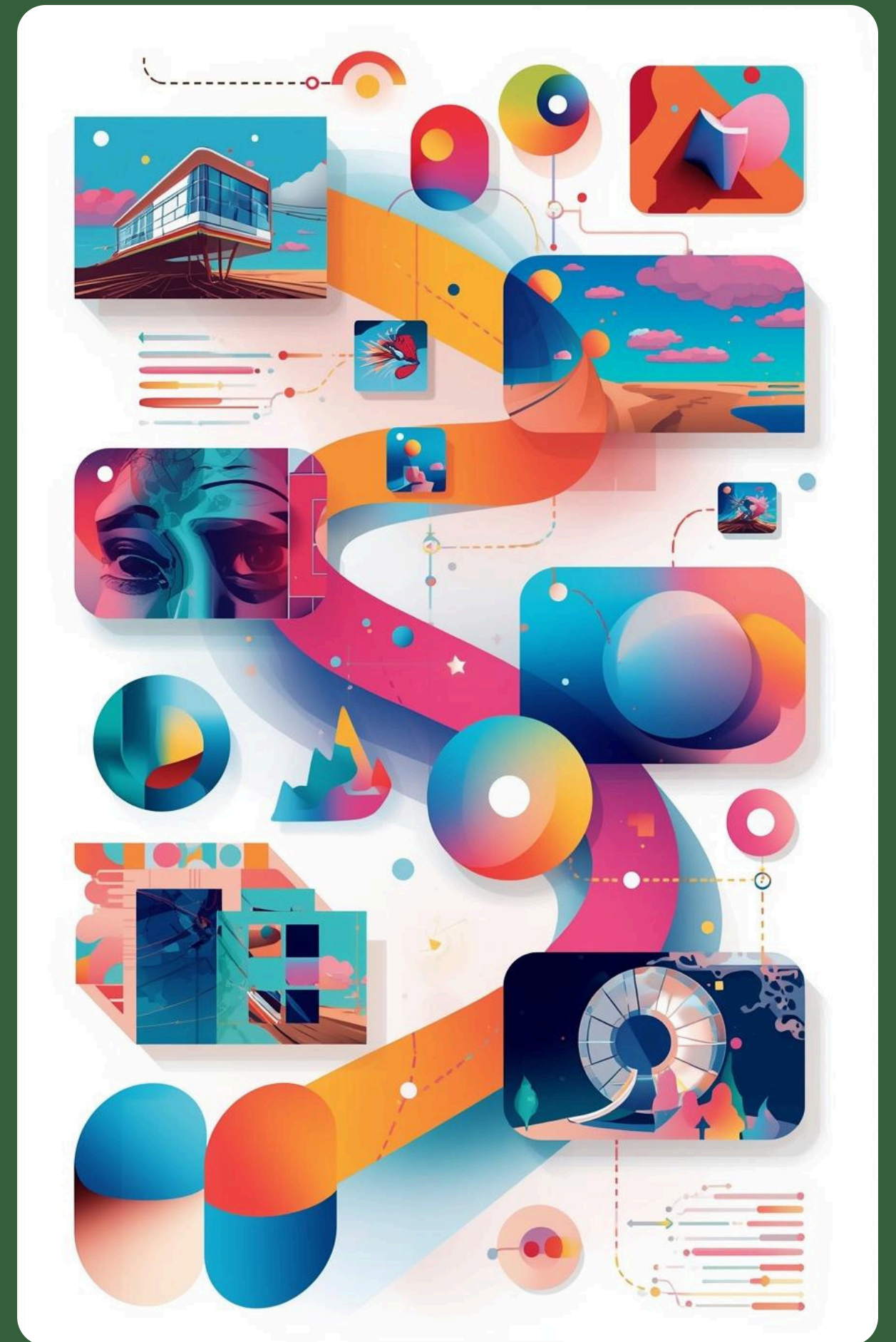


Segmentation

Masking

Text to Image

Anbu



Understanding Image Processing

Key Techniques and Their Applications



Importance of Image Processing

Computer Vision



Computer vision techniques enable machines to interpret and understand visual data effectively.

Graphics



Graphic design leverages image processing for creating stunning visuals and enhancing user experiences.

Artificial Intelligence



AI utilizes image processing techniques to improve learning capabilities and automate complex tasks.

Real-World Applications

Exploring the Impact of Image Processing

Medical Imaging

Image processing techniques like segmentation aid in **diagnosing diseases** by highlighting tumors and abnormalities, enabling healthcare professionals to make informed decisions based on accurate visual data.

Autonomous Vehicles

In autonomous driving, image segmentation and colour masking help vehicles recognize **road signs**, **pedestrians**, and obstacles effectively, ensuring safety and efficiency in navigation through complex environments.

Graphic Design

Graphic designers utilize colour masking and shape inversion to create stunning visuals in **advertising and branding**, allowing for innovative compositions that capture audience attention while conveying messages effectively.



Image Segmentation Techniques

Dividing images into meaningful regions

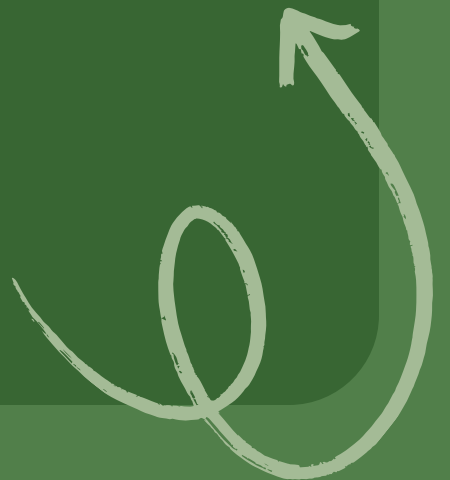


Image Segmentation Overview

01

Definition

Process of dividing images into meaningful regions.

02

Purpose

Simplifies representation for analysis and comprehension.

03

Techniques

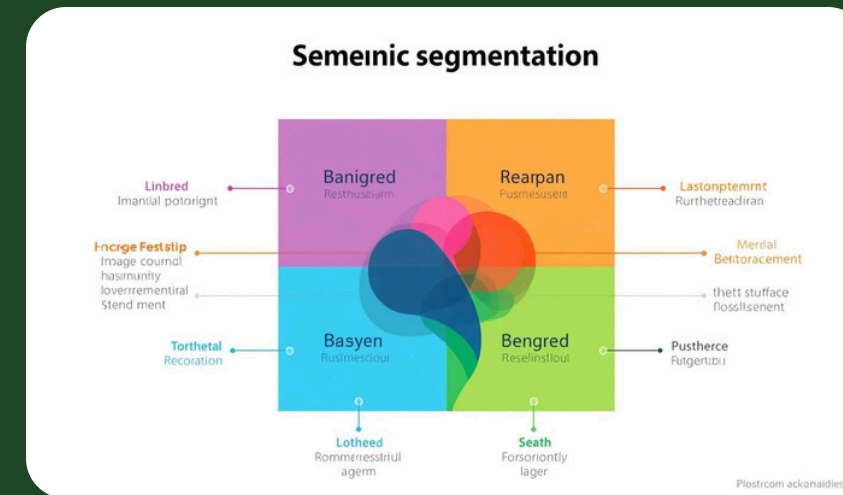
Includes thresholding, clustering, and edge detection methods.

04

Applications

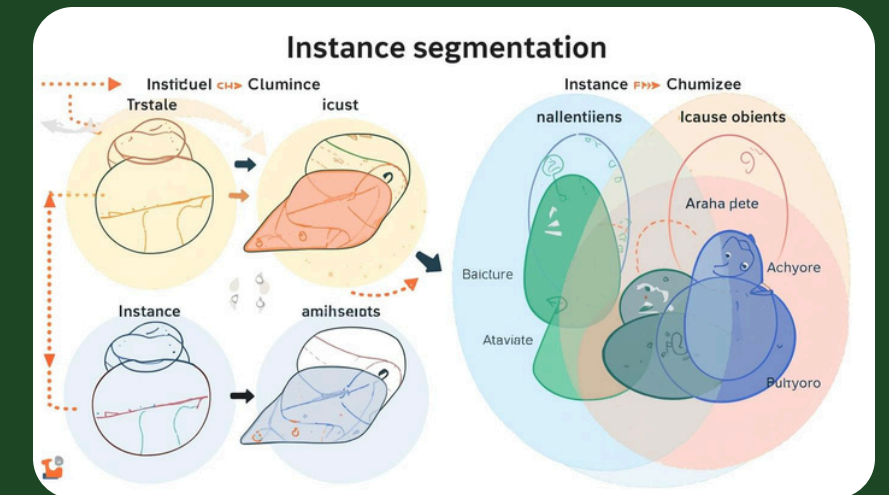
Used in medical imaging and object detection tasks.

Types of Image Segmentation Techniques



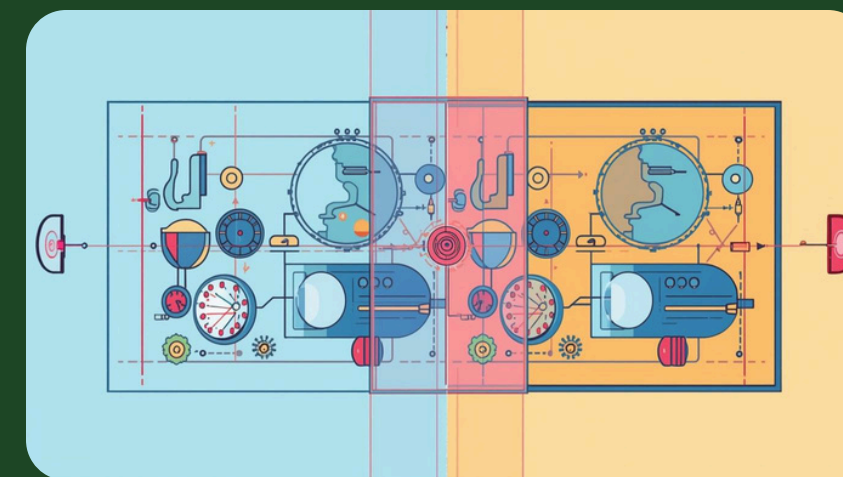
Semantic

Assigns labels to segments in an image.



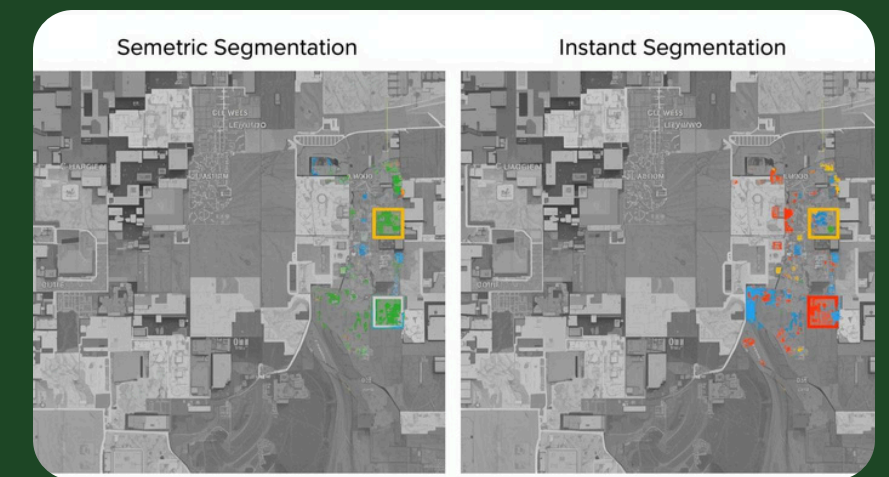
Instance

Identifies separate objects within the same category.



Panoptic

Combines semantic and instance segmentation techniques.



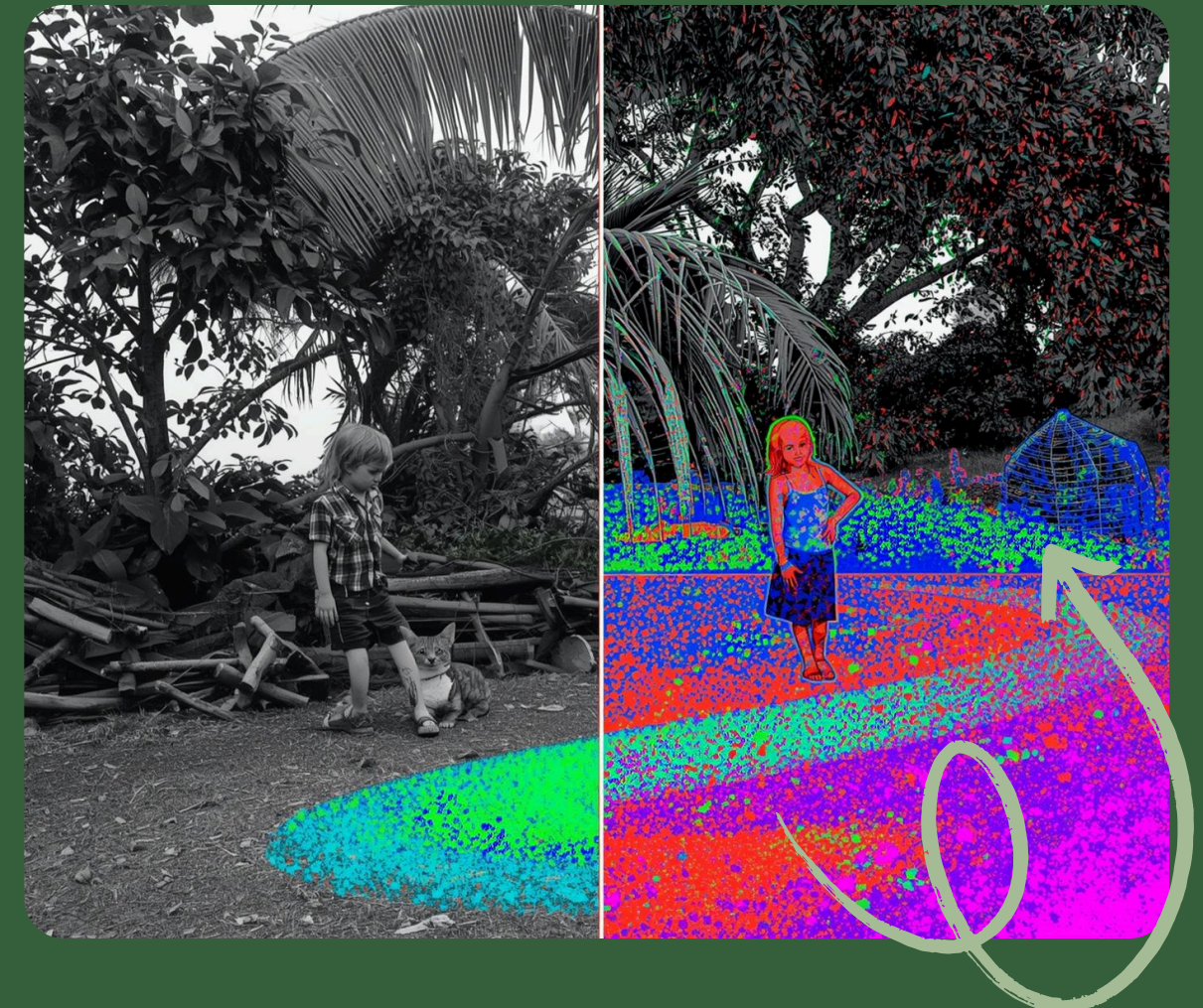
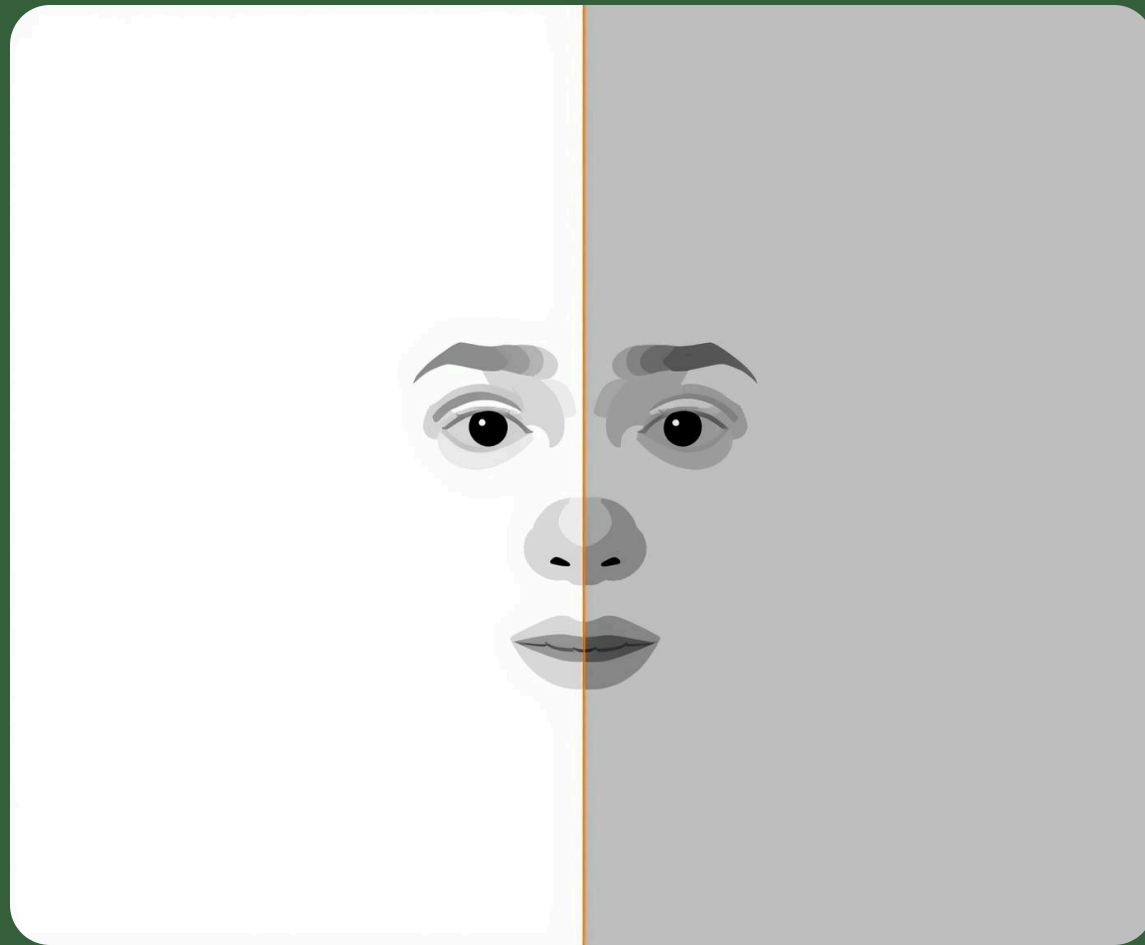
Comparison

Highlights differences between segmentation types visually.



Image Segmentation Techniques Overview

Image segmentation involves separating an image into distinct regions. Key techniques like **thresholding**, **clustering**, and **edge detection** enhance analysis in various fields, from medical imaging to autonomous vehicles.



Real-World Segmentation Examples

Tumor Detection

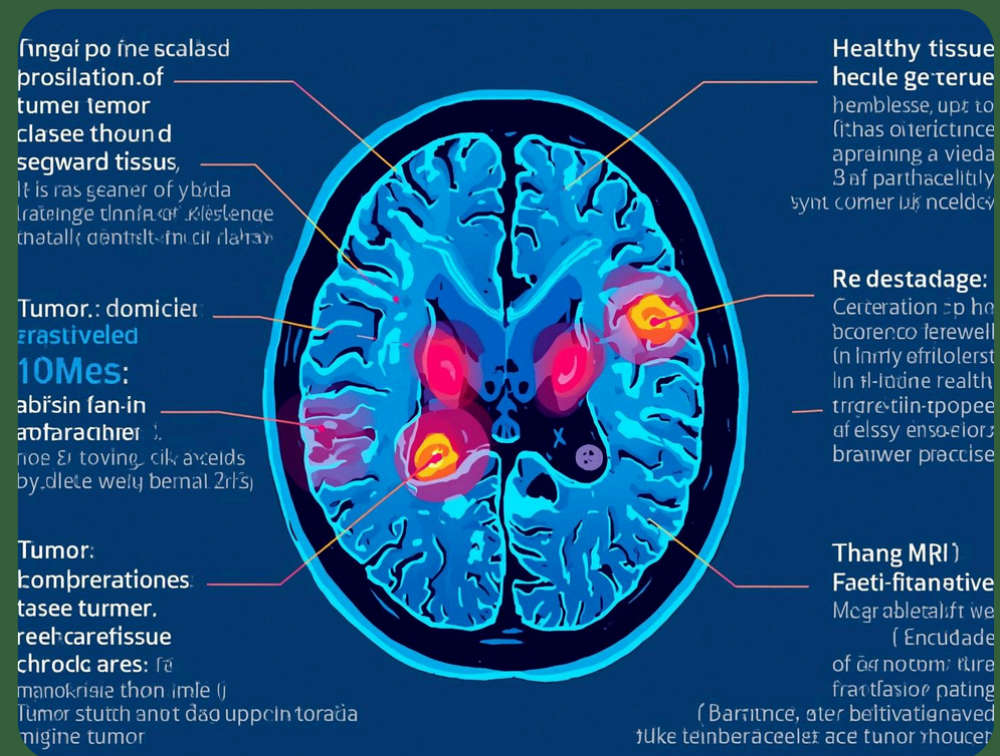
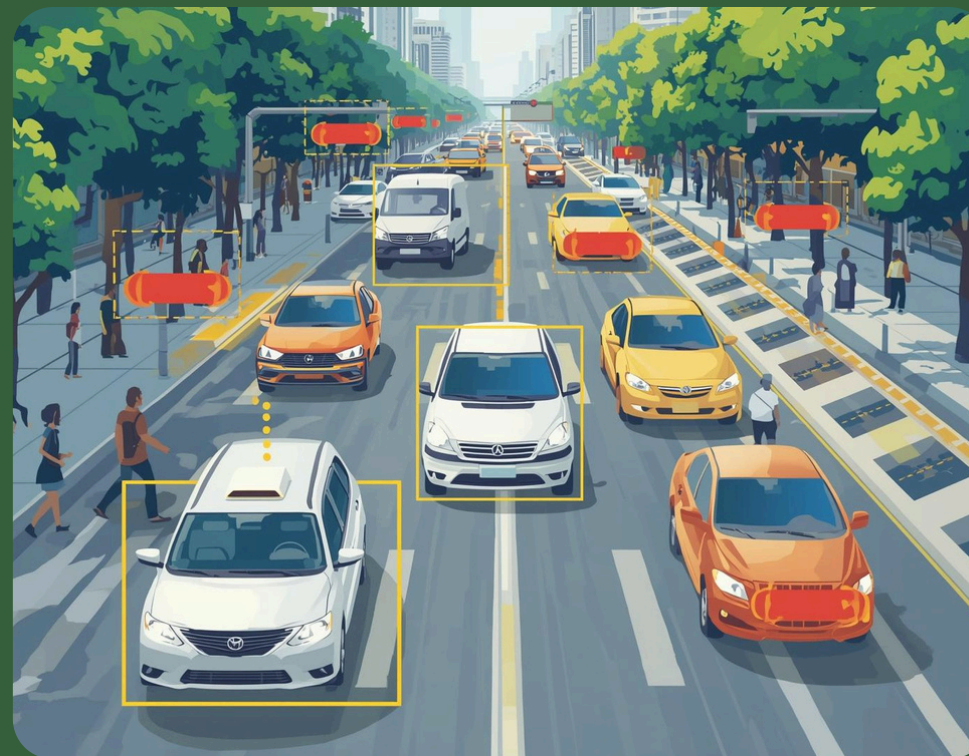


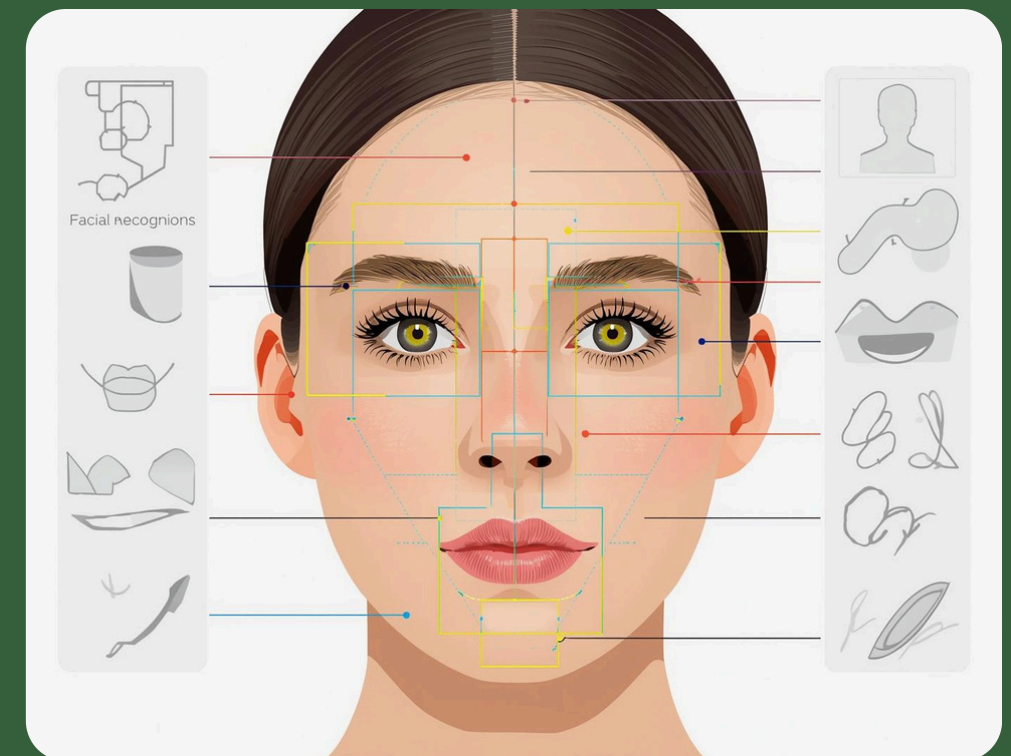
Image segmentation helps in identifying **tumor locations** in medical imaging for accurate diagnosis.

Object Detection



Effective segmentation allows autonomous vehicles to detect **pedestrians and obstacles** for safe navigation.

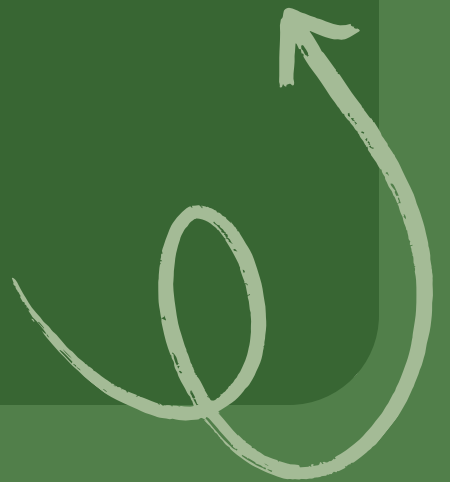
Facial Recognition



Facial recognition technology utilizes segmentation to accurately **identify and analyze faces** in images.

Colour Masking Techniques

Isolating regions based on color data



Colour Masking Overview

01

Definition

Technique to extract image parts based on color.

03

Process Steps

Involves choosing color ranges and creating binary masks.

02

Colour Spaces

RGB, HSV, and LAB are commonly used color spaces.

04

Real-World Examples

Applications include detecting ripe fruits and photo editing.

Colour Spaces

Understanding RGB, HSV, and LAB

RGB

The **RGB color model** combines red, green, and blue light in various intensities to create a wide range of colors, widely used in digital displays and imaging.

HSV

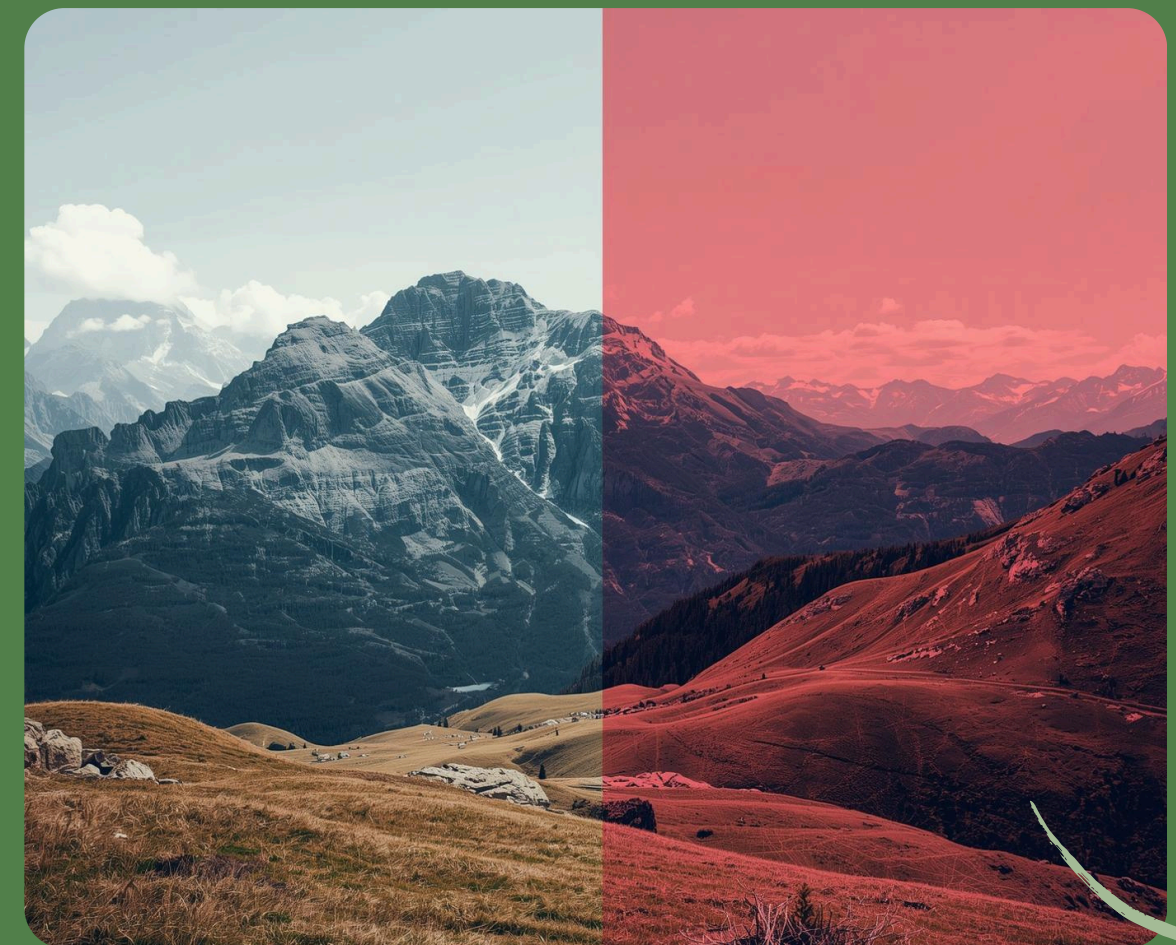
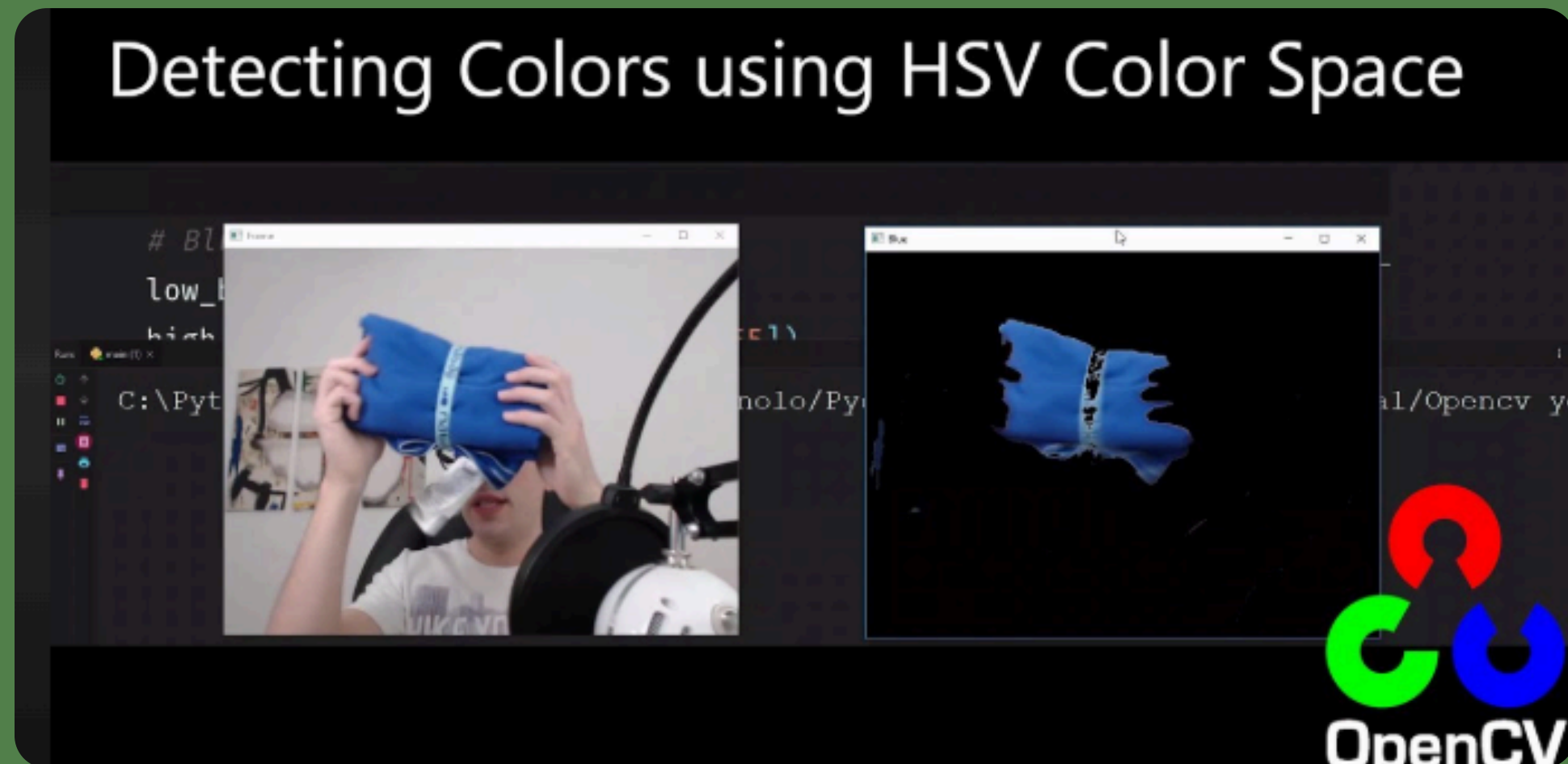
The **HSV color model** represents colors using hue, saturation, and value, making it easier for users to select and manipulate colors based on human perception.



Steps in Colour Masking Process

Overview of Creating Effective Masks

Colour masking involves three main steps: choosing appropriate colour ranges, creating a binary mask, and applying this mask to isolate desired image regions for analysis or enhancement.



Real-World Masking Examples

Detecting Ripe Fruits



This technique enhances agricultural efficiency by identifying **ripe fruits** ready for harvest.

Removing Background



Background removal simplifies **photo editing** for professional and personal use, improving focus on the subject.

Traffic Sign Recognition



This application aids in the development of **autonomous vehicles**, enhancing safety and navigation accuracy.

Otsu's Thresholding

Automatic threshold selection for image binarization



Otsu's Thresholding Explained

01

Definition

Automatically determines an optimal threshold value.

02

Minimization

Minimizes intra-class variance of pixel intensities.

03

Segmentation

Facilitates image binarization for further analysis.

04

Applications

Widely used in various computer vision tasks.

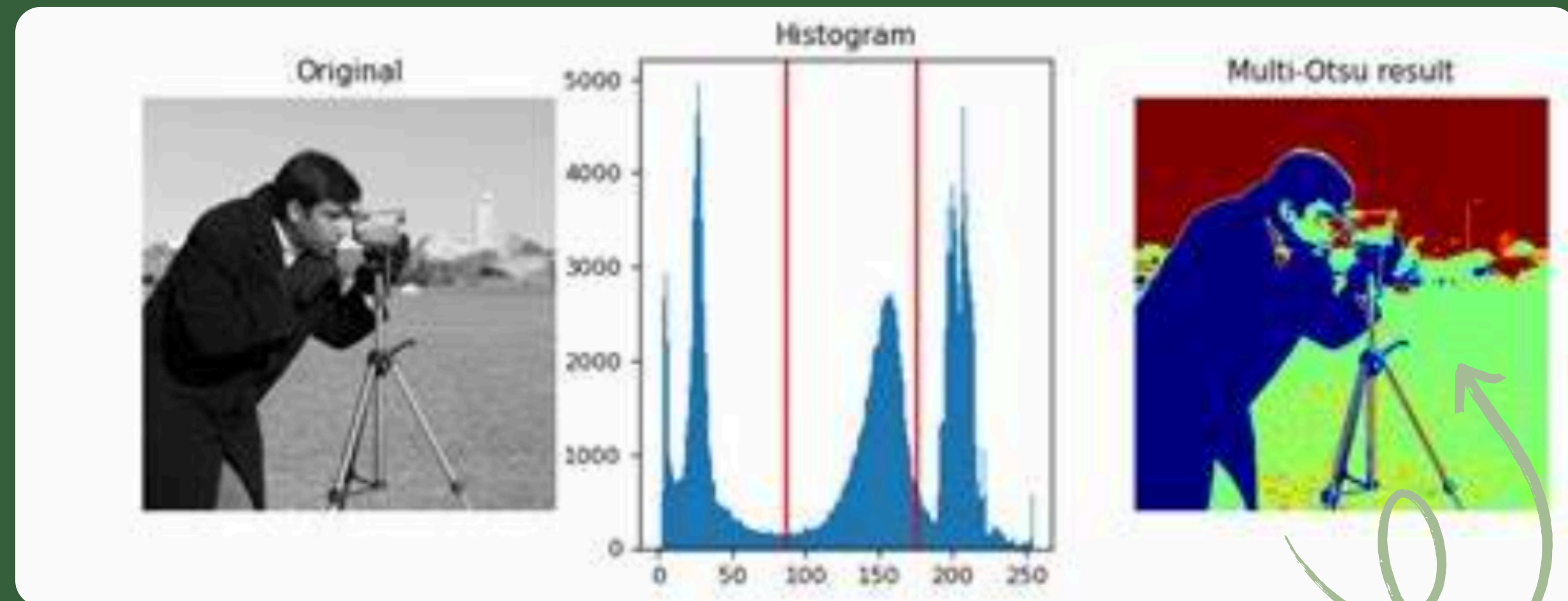
Real-World Applications of Thresholding

Document Scanning and Defect Detection

Thresholding techniques play a crucial role in **document scanning**, **license plate recognition**, and defect detection, enabling efficient analysis and clarity in image processing tasks.



An example image thresholded using Otsu's algorithm



Summary

Image Segmentation

The applications of image segmentation are in many fields, including medical imaging, computer vision, and robotics.

Image Segmentation

Image Segmentation

Image Segmentation

Image Segmentation

Colour Masking

The applications of colour masking are in many fields, including medical imaging, computer vision, and robotics.

Colour Masking

Colour Masking

Colour Masking

Colour Masking

Eyetracking

The applications of eyetracking are in many fields, including medical imaging, computer vision, and robotics.

Eyetracking

Eyetracking

Eyetracking

Eyetracking

Recap of image processing techniques